Universitat de Girona

Robotic Manipulation

IFROS/MIRS

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Assistant researcher

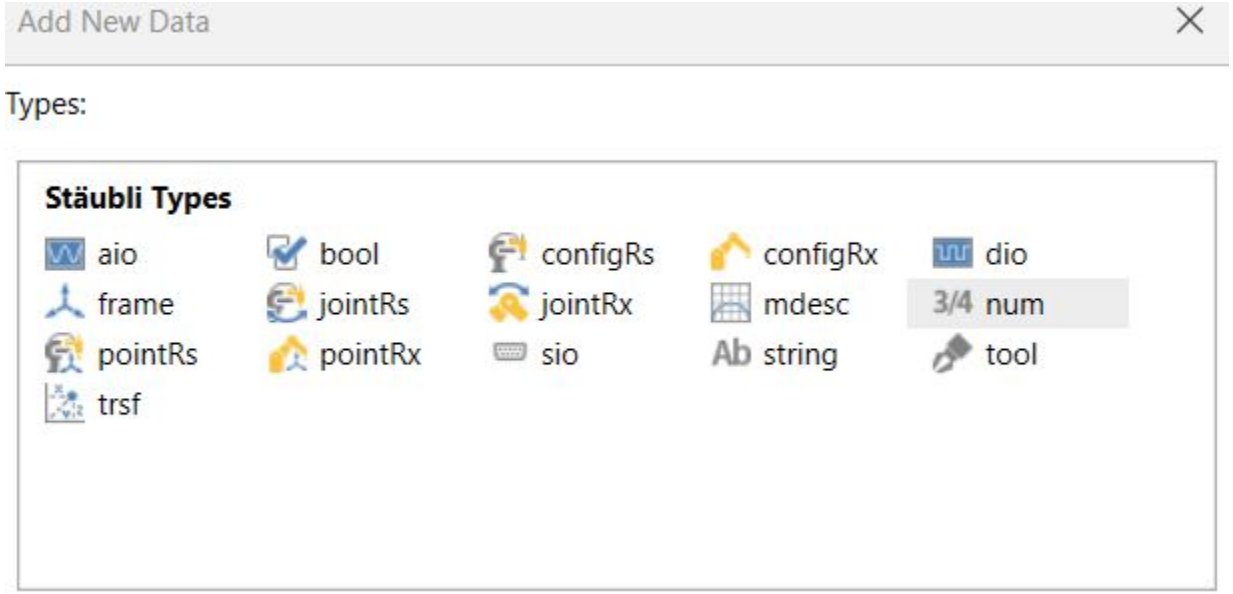
VICOROB

2025

UNDERWATER VISION and ROBOTICS LAB

Lab1 Overview - Parameters setup

Set up some data types for our application



Lab1 Overview - Parameters setup

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Vector of desired joint values



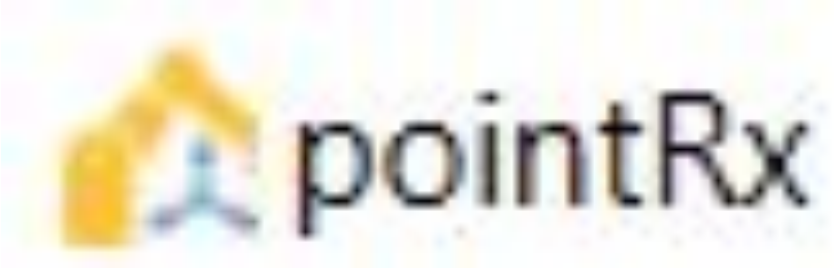
testapp-position0 Data* | testapp-start | testapp-stop | 3D View

Index	J1	J2	J3	J4	J5	J6
0	0	0	0	0	0	0

```
movej(position0, tool0, mNomSpeed)
```

Lab1 Overview - Parameters setup

Set of cartesian coordinates (X,Y,Z,RX,RY,RZ)



testapp-position2 Data | testapp-start | testapp-stop | 3D View

Index	X	Y	Z	Rx	Ry	Rz	Shoulder	Elbow	Wrist
0	379,5	-289,5	-285,18	-180	0	90	sfree	efree	wfree
1	546,99	-61	-256,18	180	0	-90	sfree	efree	wfree

```
movej(position2[0], tool0[0], mNomSpeed)
```

Best advice: Check the manual

STÄUBLI

9.6. MOVEMENT INSTRUCTIONS

void **movej**(joint jPosition, tool tTool, mdesc mDesc)

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void **movej**(point pPosition, tool tTool, mdesc mDesc)

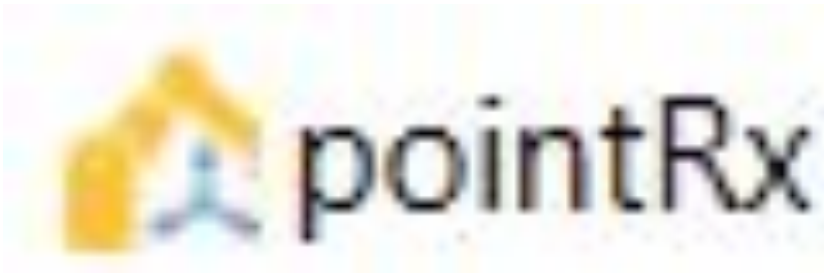
---

*Records a command for a joint movement towards the **pPosition** or **jPosition** positions, using the **tTool** and the **mDesc** movement parameters.*



Lab1 Overview - Parameters setup

Set of cartesian coordinates (X,Y,Z,RX,RY,RZ)



D View

	Rz	Shoulder	Elbow	Wrist
0	90	sfree	efree	wfree
0	-90	sfree	efree	wfree

Arm joint link configuration (shoulder, elbow, wrist)

Arm Configurations for point P

8.7. CONFIG TYPE

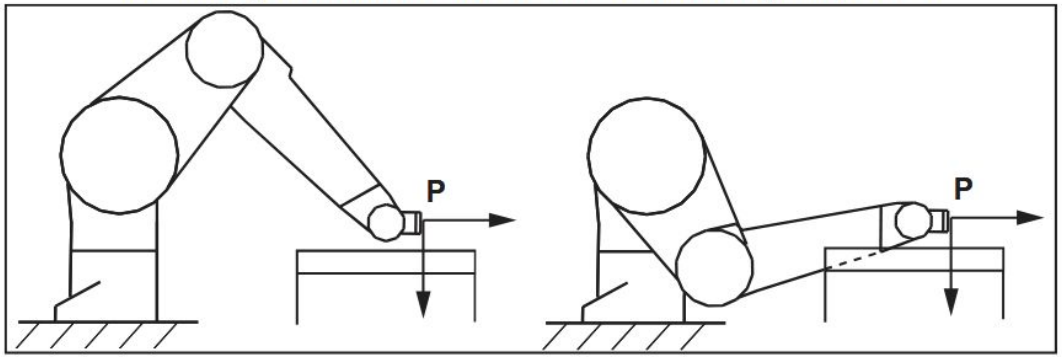
The configuration concept of a Cartesian point is an "advanced" concept; this chapter can be skipped the first time you read this document.

8.7.1. INTRODUCTION

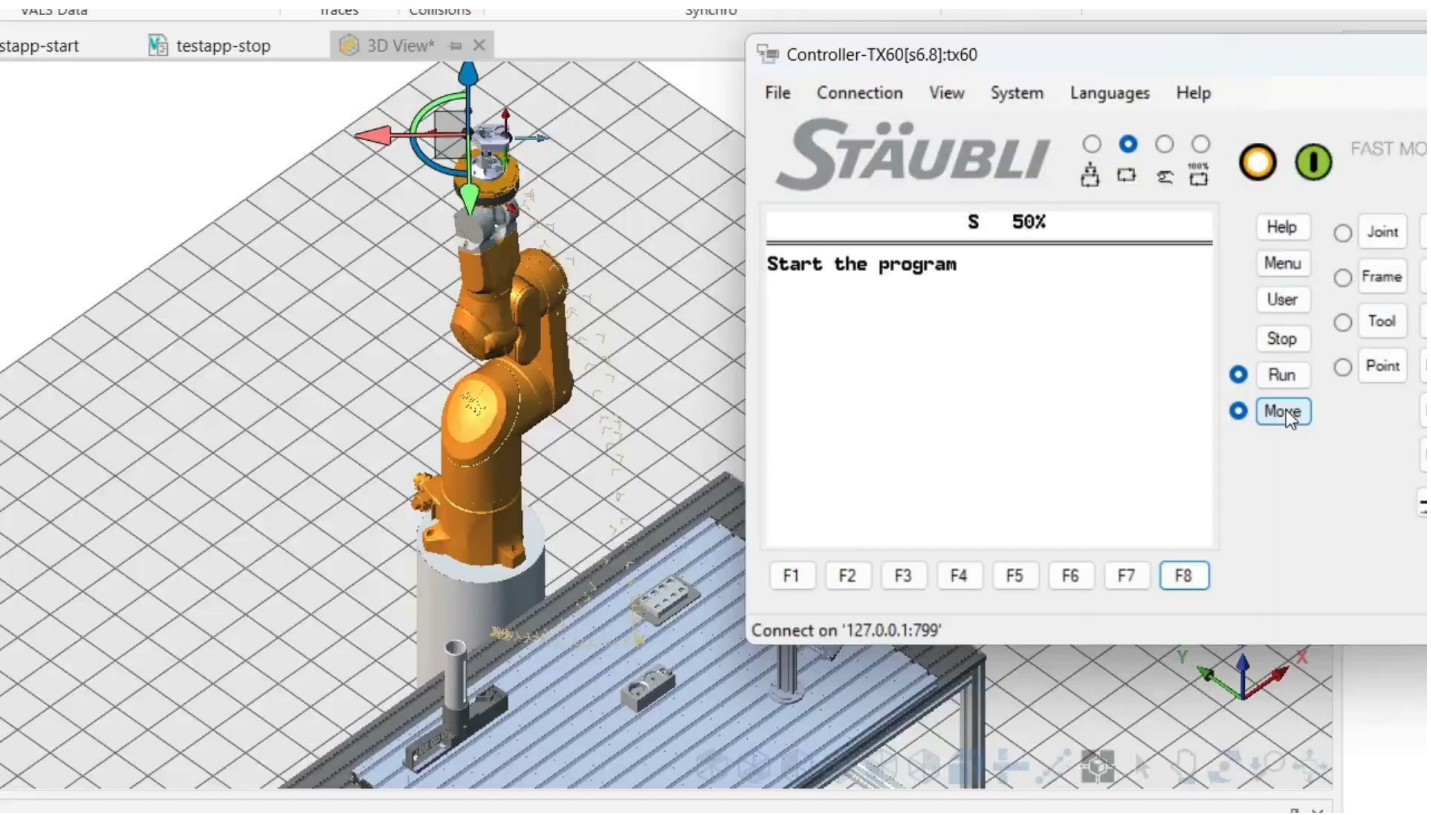
There are generally several ways in which a robot can reach a given Cartesian point.

These possibilities are known as "configurations". The figure below illustrates two different configurations:

Two configurations that can be used to reach a given point: P



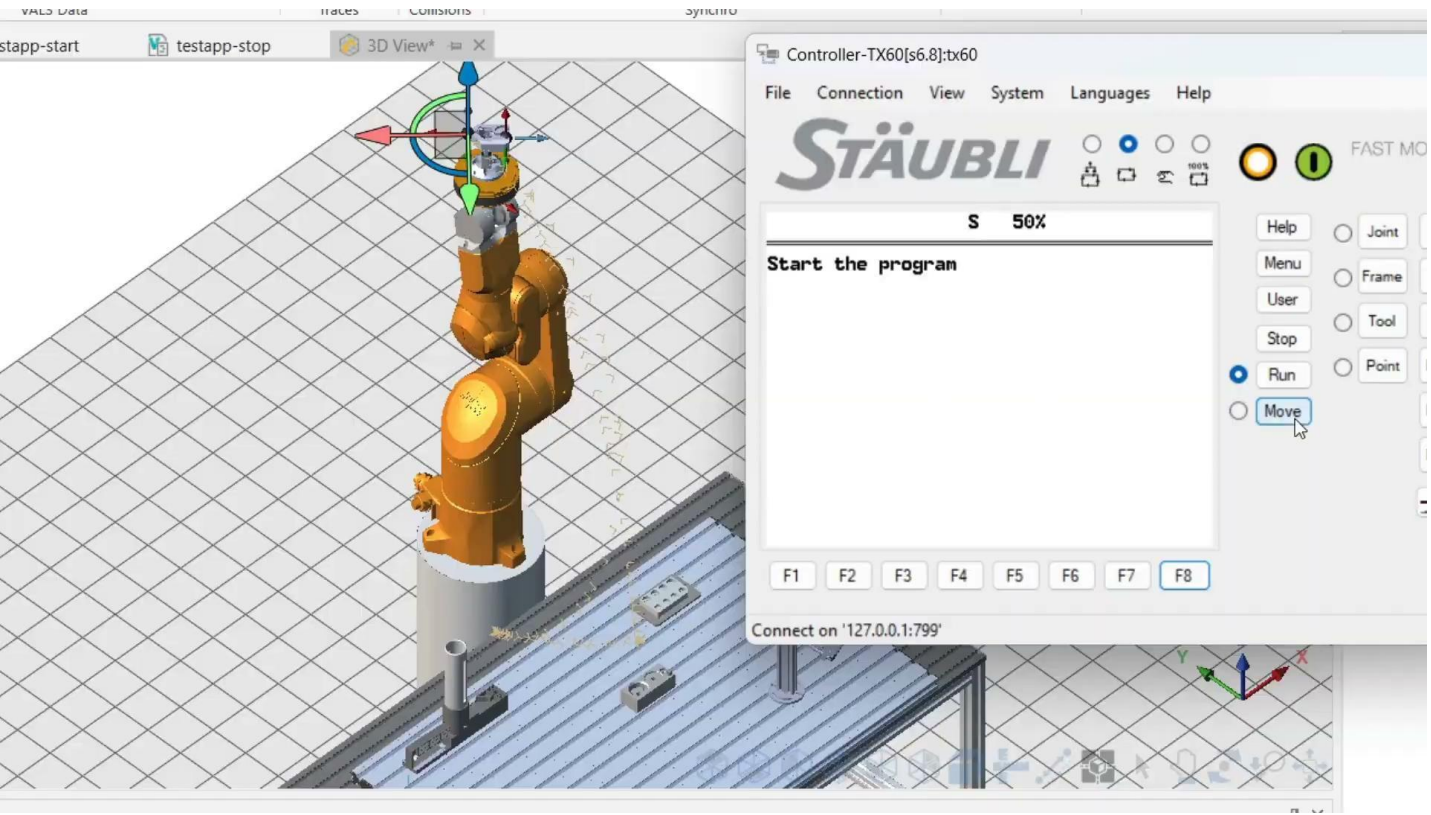
Arm Configurations for point P



point	shoulder
P0	lefty
P1	lefty

Shoulder joint has minimal travel

Arm Configurations for point P



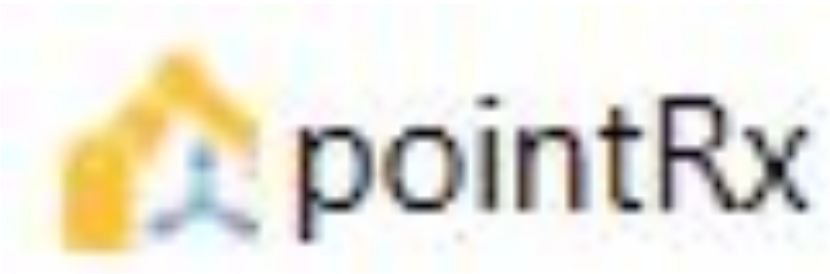
point	shoulder
P0	rigthty
P1	lefty

Shoulder joint has *crazy* travel

Lab1 Overview - Parameters setup

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Set of cartesian coordinates (X,Y,Z,RX,RY,RZ)



D View

	Rz	Shoulder	Elbow	Wrist
0	90	sfree	efree	wfree
0	-90	sfree	efree	wfree

Assume kinematics controller will choose whatever solutions finds first (**NOT SAFE**)

Lab1 Overview - Application code

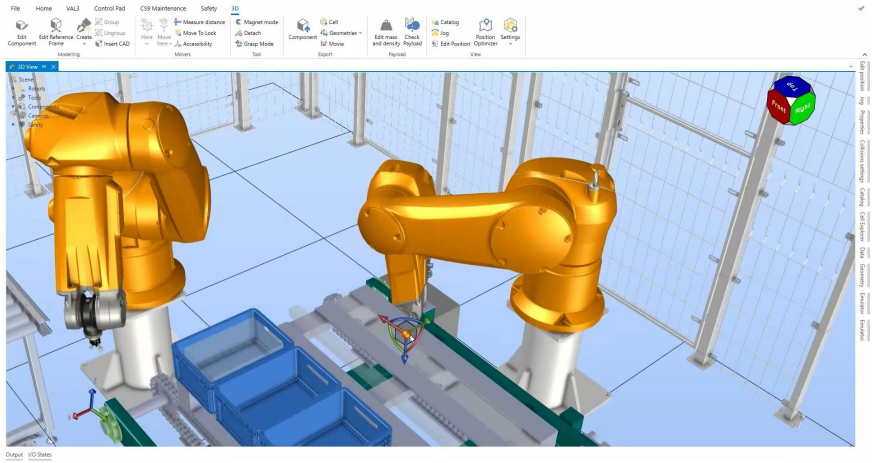
```
begin
  userPage()
  cls()
  putln("Start the program")
  enablePower()
  mNomSpeed.vel=30
  movej(position2[0], tool0[0], mNomSpeed)
  close(tool0[0])
  putln("Position 2a reached. Terminal element closed.")
  open(tool0)
  movej(position2[1], tool0[1], mNomSpeed)
  close(tool0[1])
  putln("Position 2b reached. Terminal element closed.")
end
```



Lab Work Flow

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Everything validated first in simulation



Then upload and execution in real robot



Deliverables

- Video of Simulation
- Video of Real execution

{ 70%

- Lab Report
- Code files

{ 30%

Lab Report (industry-like)

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Like robot programmers in a factory:

1. **Task Description** – e.g., “Pick part from A, place in B.”
2. **Program Breakdown** – Motion sequence (MoveJ, MoveL, etc.), I/O operations, waits.
   a. *Flow diagram or pseudocode*
3. **Parameters Used** – Speeds, tool frames, payload.
4. **Safety Considerations** – Collisions avoided, safe home position.
5. **Execution Results** – Did the sequence complete? Cycle time?
6. **Improvements** – how would you improve or optimize the task.